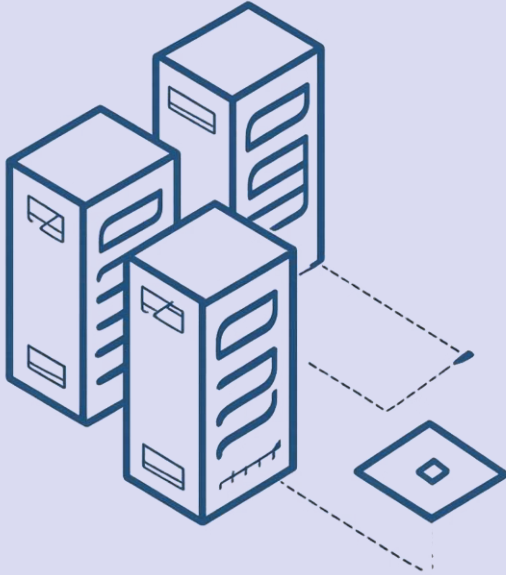




Data Warehouse Architecture and Design Process

Data Mining and Data Warehousing (DMW)

Why Do We Need Data Warehouse Architecture?

Modern enterprises generate massive data across many systems. A data warehouse centralizes this data for fast, reliable analytics.



Multiple Heterogeneous Sources

Operational DBs, ERP, CRM, spreadsheets, and APIs



Centralized Storage

One source of truth for the entire organization



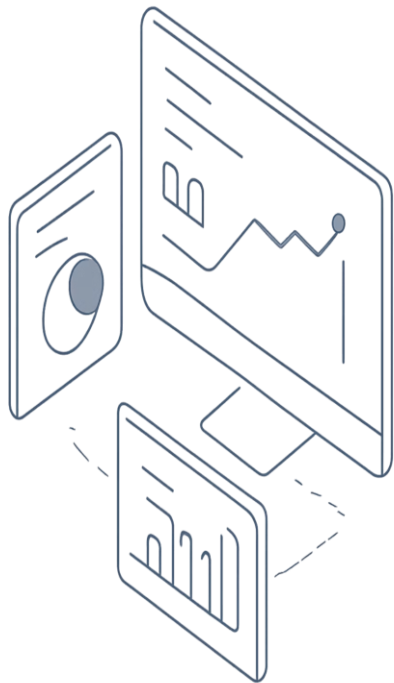
Fast Decision Making

Optimized for analytical queries, not transactions



Scalability

Handles growing volumes without performance loss

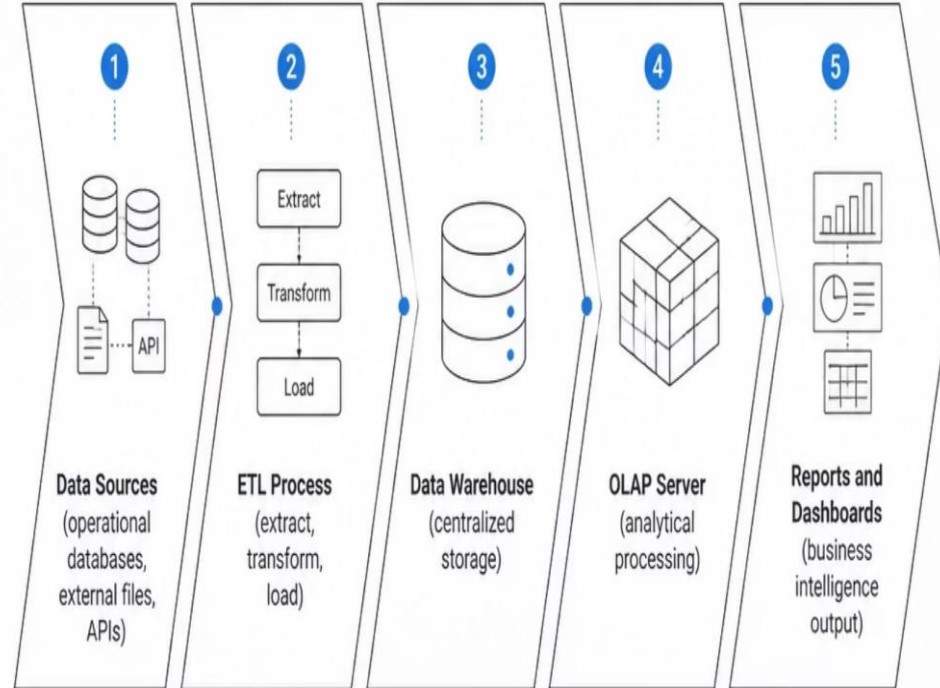


❏ **Real-World Example:** Amazon collects orders → operational DBs → central warehouse → analytics dashboard.

Definition of Data Warehouse Architecture

A **data warehouse architecture** is the logical and physical organization of all components that collect, store, process, and deliver data for analysis.

- **Logical structure:** how components relate and communicate
- **Physical structure:** servers, storage, and network layout
- **Data movement:** defined flow from sources to end users
- **Communication:** protocols and interfaces between layers



Definition of Data Warehouse Architecture

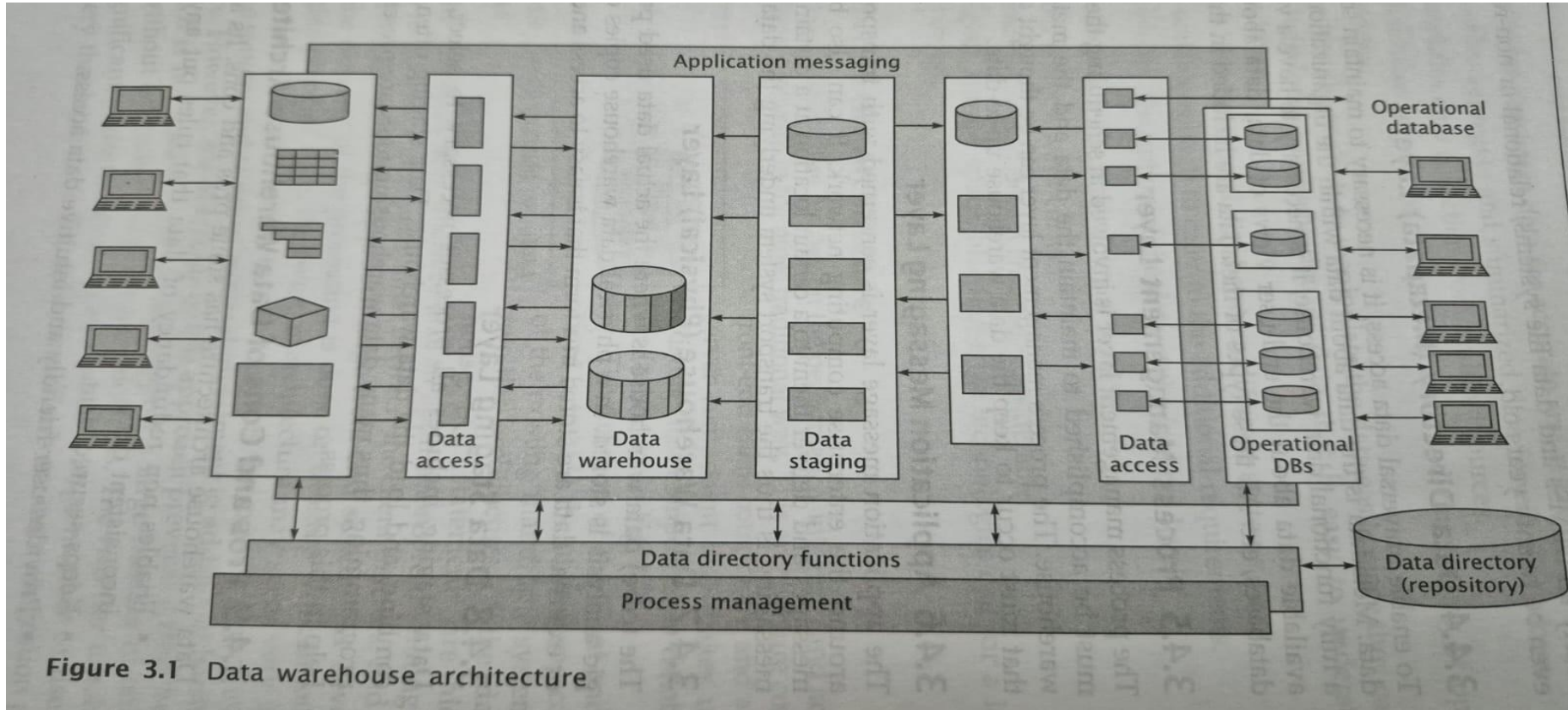
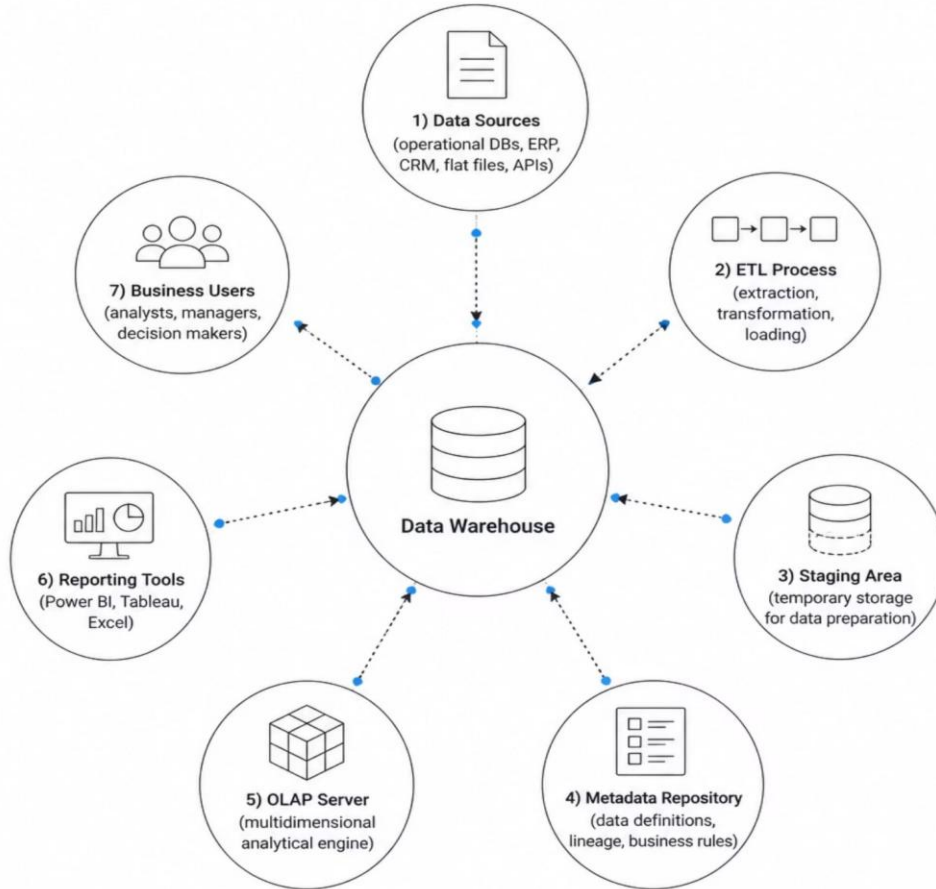


Figure 3.1 Data warehouse architecture

Components of Data Warehouse Architecture

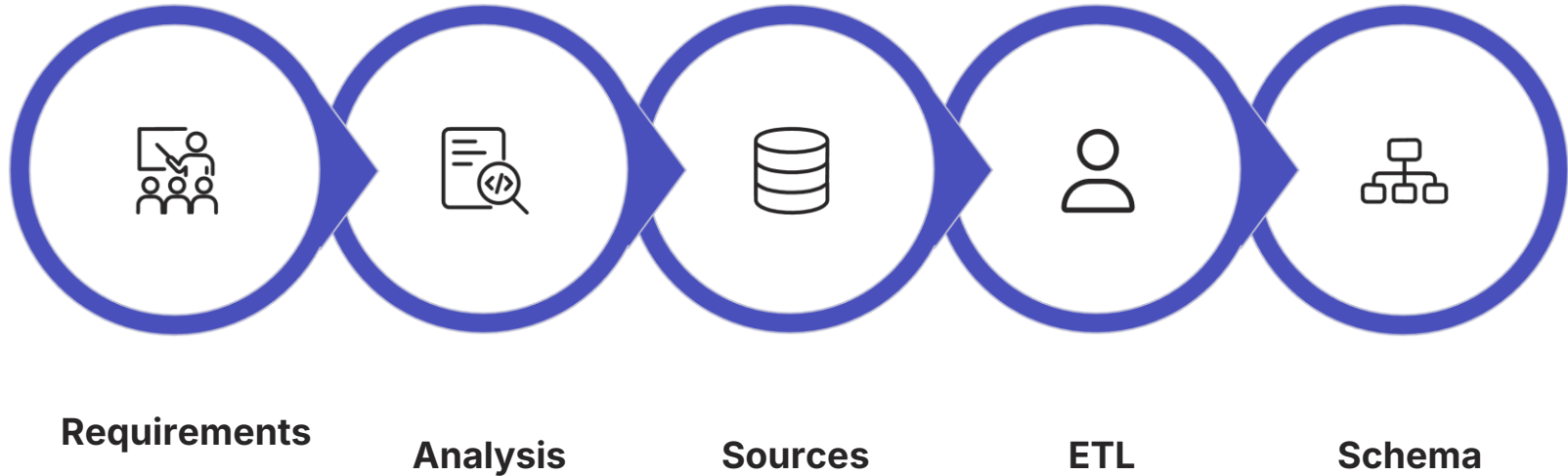


Each component plays a distinct role in the data pipeline from source systems to decision makers.

- **Data Sources:** operational DBs, ERP, CRM, flat files, APIs
- **ETL Process:** extracts, transforms, and loads data
- **Staging Area:** temporary zone for data preparation
- **Metadata Repository:** stores definitions, lineage, and rules
- **OLAP Server:** enables multidimensional analytical queries
- **Reporting Tools:** Power BI, Tableau, Excel for visualization
- **Business Users:** analysts and managers consuming insights

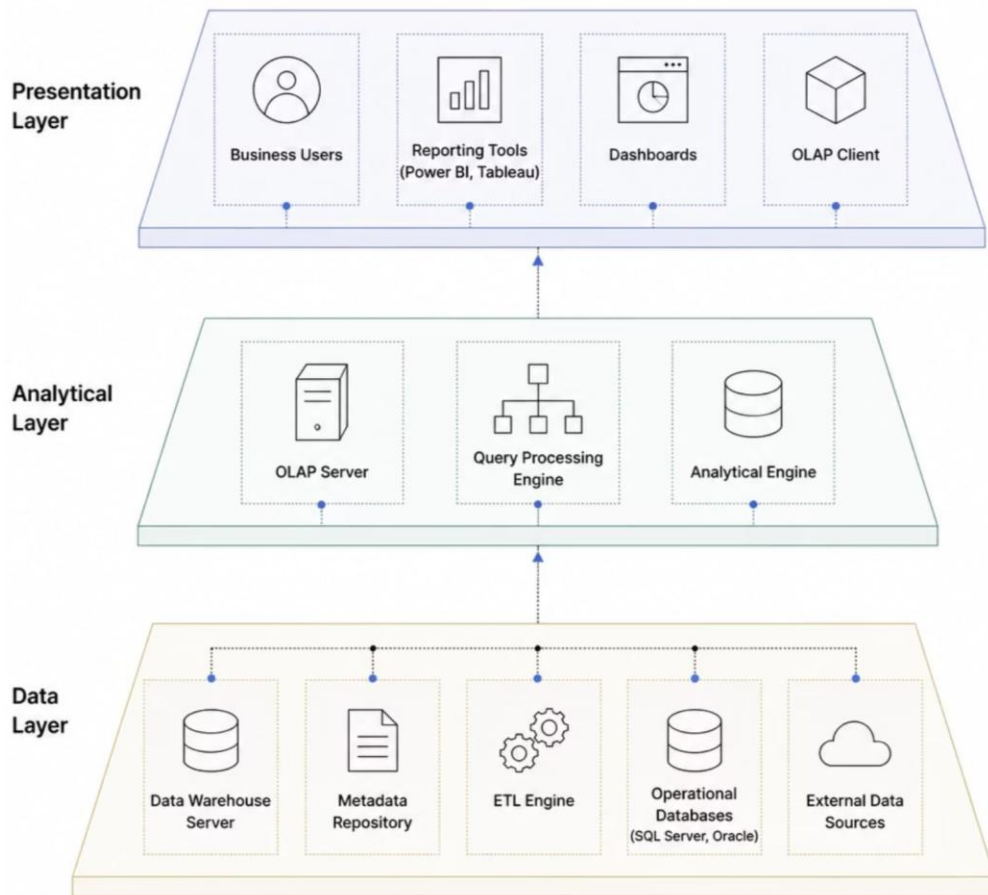
Data Warehouse Design Process

A structured, step-by-step methodology ensures the warehouse meets business needs and remains maintainable over time.



Each stage builds on the previous one, skipping requirements or testing leads to costly rework in production.

Three-Tier Data Warehouse Architecture



The three-tier model cleanly separates **storage**, **processing**, and **presentation** — enabling independent scaling and maintenance of each layer.

- **Bottom Tier:** Data warehouse server, metadata, ETL, operational DBs, external sources
- **Middle Tier:** OLAP server, query processing, analytical engine
- **Top Tier:** Business users, reporting tools, dashboards, OLAP clients

Data flows **upward** from sources through processing to end users.

Explaining Each Tier



Bottom Tier — Data Layer

Purpose: Store and manage raw and integrated data

Components: Data warehouse server, metadata, ETL engine, operational DBs

Tools: SQL Server, Oracle, MySQL, PostgreSQL



Middle Tier — Analytical Layer

Purpose: Process complex analytical queries efficiently

Components: OLAP engine, query processor, aggregation services

Tools: SSAS, Oracle OLAP, Apache Kylin



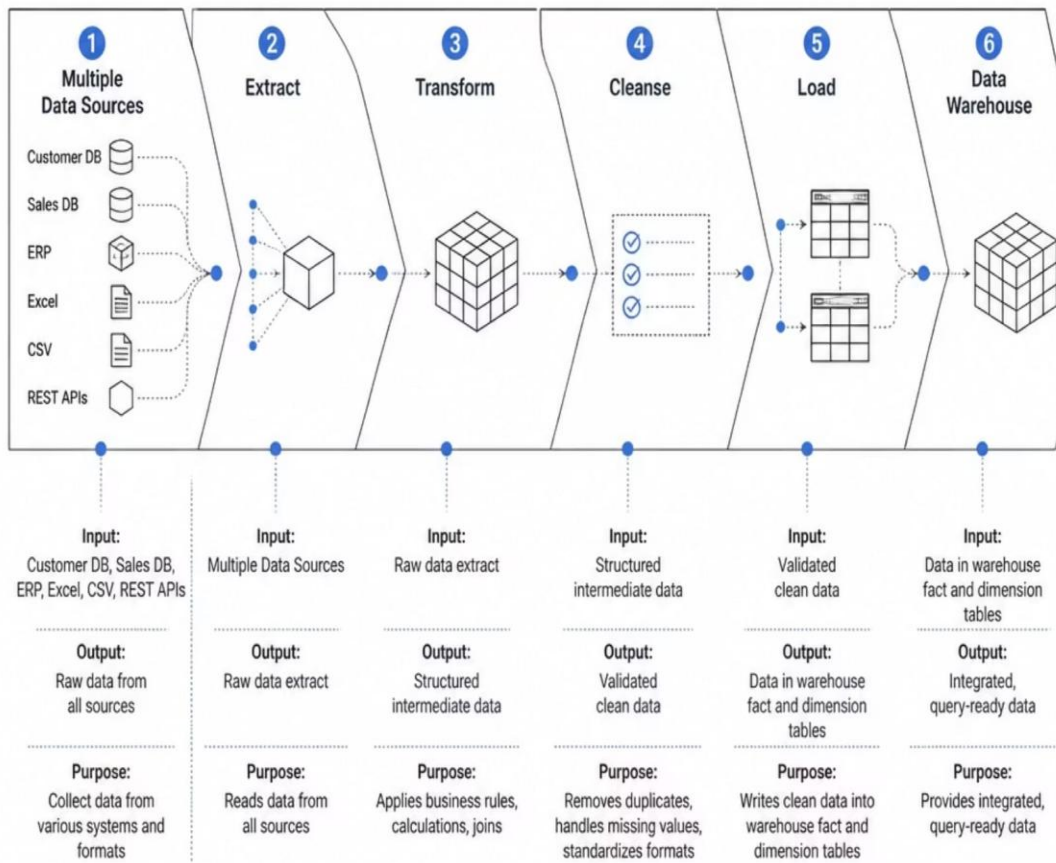
Top Tier — Presentation Layer

Purpose: Deliver insights to business users

Components: Reporting tools, dashboards, OLAP clients

Tools: Power BI, Tableau, Excel, QlikView

Back-End Tools and the ETL Pipeline



ETL (Extract, Transform, Load) is the backbone of data warehouse population — ensuring only clean, consistent data reaches the warehouse.

- **Extract:** Read from heterogeneous sources (DBs, APIs, files)
- **Transform:** Apply business rules, calculations, joins
- **Cleanse:** Handle missing values, duplicates, format errors
- **Load:** Write validated data into fact and dimension tables
- **Metadata:** Track data definitions, lineage, and transformations
- **Scheduling:** Automate ETL jobs (daily, hourly, real-time)

Data Cleansing and Transformation

Common Data Quality Issues

- **Missing Values**
Impute with defaults or flag as NULL
- **Duplicates**
Remove redundant records
- **Standardization**
Unify formats (dates, currency, units)
- **Validation**
Enforce business rules and constraints

Before → After Cleansing

Issue	Before	After
Missing Value	NULL	0 or "Unknown"
Duplicate	Two identical rows	One row retained
Date Format	01/02/2024	2024-02-01
Currency	\$1,200 / 1200USD	1200.00 USD
Category	"elec", "Electronics"	"Electronics"

☐ Clean data is the foundation of reliable analytics.

Industry Case Study: Flipkart's Data Warehouse

Flipkart processes millions of daily transactions across sales, inventory, payments, and customer interactions — all feeding a centralized analytics platform.



Data Sources

Sales, Customers, Inventory, Payments, Returns, Marketing



ETL Pipeline

Extract, cleanse, transform, and load daily batches



Central Warehouse

Integrated, query-ready data store



Power BI Dashboard

Real-time KPIs and reports for management



Key Takeaways: Architecture defines data flow from sources to decision makers.

Three-tier design separates storage, processing, and presentation. ETL ensures data quality. A well-designed warehouse enables faster, better decisions.

